Pratyush Prakash

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.10
Manava Bharti Heritage School	Class XII (CBSE)	2022	65.20%
DAV Public School	Class X (CBSE)	2020	73.40%

EXPERIENCE

CyanConnode Pvt. Ltd. , *AI/ML Intern*

- Built and tested AI-driven workflows using **Python, LangChain, and RAG** (Retrieval-Augmented Generation), applying LLM and Agentic AI techniques to real-world IoT use cases.
- Analyzed IoT-generated datasets using **SQL and Python**, extracting insights to support engineering and product decisions.
- Collaborated using **Git and GitHub** in a distributed team, contributing to Generative AI and Machine Learning experiments within an IoT product environment

Estate Plus CRM ,*Business Development Intern*

- **Generated** and qualified **B2B leads** through outbound calls and follow-ups, contributing directly to the sales pipeline for a CRM software product.
- **Managed** customer interaction records and sales funnel stages within the CRM, improving lead-tracking accuracy for the sales team.
- **Presented** product solutions to prospective clients and coordinated demos, strengthening client communication and negotiation skills.

PROJECTS

Performance Evaluation of a Microbial Fuel Cell with Microalgae-Assisted Cathode | *IIT Hyderabad*

Semester Research Project under Dr. Pritha Chatterjee

- Designed and operated an H-type Microbial Fuel Cell (MFC) with a microalgae-assisted cathode, achieving **83.33% COD removal** over 4 batch cycles.
- Generated stable bio-electricity of **600-700 mV** using a Nafion 117 membrane and anaerobic mixed-culture inoculum.
- **Monitored** voltage, current, pH, and COD daily for **30+ days**, evaluating microalgae-assisted cathodic oxygen supply and system performance.
- **Analyzed** biofilm formation and electron transfer, demonstrating the potential of MFCs for sustainable wastewater treatment and aeration-energy reduction; co-authored and presented findings as a poster.

Enterprise Document Intelligence & Multi-Agent RAG

Python, LangChain, LangGraph, FastAPI, LLM, ChromaDB

- Built a **RAG-based document intelligence system** to retrieve and analyze relevant information from large PDF collections.
- Implemented **document chunking, embeddings, and ChromaDB vector search** for efficient semantic retrieval.
- Developed a **LangGraph multi-agent workflow** with specialized agents for document retrieval, analysis, and response generation.
- Integrated **LLM APIs with source-grounded prompting** to generate context-aware responses and reduce irrelevant information.

POSITIONS OF RESPONSIBILITY

Hostel Representative(HR), *Student Gymkhana, IIT Hyderabad*

- Elected student representative handling HR-related hostel matters and administration liaison.
- Identified recurring student grievances and strengthened communication channels between residents and hostel

administration.
- Improved hostel-administration responsiveness by consistently escalating and resolving HR concerns.

Coordinator (Intern), Operations, *Inter-IIT Sports Meet 2025-26*

- Operations coordinator for a national-level multi-institute sporting event.
- Reviewed event logistics requirements across teams and coordinated on-ground plans to ensure smooth scheduling.
- Delivered seamless event execution, contributing to its successful organization as recognized by the Operations Head.

SKILLS

AI/ML: Machine Learning, Generative AI, LLMs, Agentic AI, NLP, RAG, Prompt Engineering

Soft Skills: Problem Solving, Analytical Thinking, Communication, Teamwork

Libraries & Frameworks: LangChain, Scikit-learn, Pandas, NumPy, OpenCV

RELEVANT COURSES

Extra Courses: Spoken Japanese, Introduction to Entrepreneurship,Communication Skills, Science Technology & Society, What's on the Menu of Food ond Eating,Madhubani Painting,Martial Arts

Mathematics: Calculus I & II, Elementary Linear Algebra, Differential Equations, Transform Techniques, Probability, Complex Variables, Statistics

EXTRACURRICULAR

- Represented hostel in **Inter-Hostel Sports Competitions (Milan GC)**
- Member of the **Inter-Department Cricket Team**
- **Won 2nd Prize** in Milan GC Drama Play Competition.
- **Part of NSS** (National Service Scheme) Volunteer
- **Hobbies:** Gym, Cricket
- **Interests:** Reading